

ECON2050 Tutorial 6

Linear Algebra II

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Today's Plan

- ▶ Tutorial questions: Q1–Q4

Tutorial Question 1

Problem 1

Let

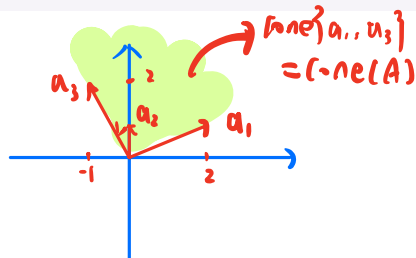
$$\mathbf{A} = \begin{pmatrix} 2 & 0 & -1 \\ 1 & 1 & 2 \end{pmatrix}.$$

- (i) Sketch the cone generated by the columns of $\mathbf{A}$.
- (ii) What is the cone generated by just the first and third columns of $\mathbf{A}$?
- (iii) If $\mathbf{b} = (1, 0)$, decide whether $\mathbf{Ax} = \mathbf{b}$ has a solution with $\mathbf{x} \geq 0$.

$$(i) \mathbf{A} = \begin{pmatrix} 2 & 0 & -1 \\ 1 & 1 & 2 \end{pmatrix} = (\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3) \quad \left\{ \begin{array}{l} \mathbf{a}_1 = (2, 1) \\ \mathbf{a}_2 = (0, 1) \\ \mathbf{a}_3 = (-1, 2) \end{array} \right.$$

$$\text{cone}(\mathbf{A}) = \{x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + x_3\mathbf{a}_3 : x_1, x_2, x_3 \in \mathbb{R}_+\}$$

The cone is the region between the rays generated by $\mathbf{a}_1 = (2, 1)$ and $\mathbf{a}_3 = (-1, 2)$



(ii)

$$a_2 = x_1 a_1 + x_3 a_3$$

$$(0, 1) = x_1 \cdot (2, 1) + x_3 \cdot (-1, 2)$$

$$(0, 1) = (2x_1 - x_3, x_1 + 2x_3)$$

$$\begin{cases} 0 = 2x_1 - x_3 \\ 1 = x_1 + 2x_3 \end{cases} \Rightarrow \begin{cases} x_1 = \frac{1}{5} > 0 \\ x_3 = \frac{2}{5} > 0 \end{cases}$$

$\Rightarrow a_2$ is a non-negative linear combination of a_1 and a_3

$\Rightarrow a_2 \in \text{cone}\{a_1, a_3\}$

$\Rightarrow a_2$ is useless so $\text{cone}(A) = \text{cone}\{a_1, a_3\}$

(iii) $AX = b$ where $A = \begin{pmatrix} 2 & 0 & -1 \\ 1 & 1 & 2 \end{pmatrix}$, $b = (1, 0)$

$\Rightarrow$ whether it has a non-negative solution x ?

$F \neq \emptyset$ Solution $\exists$ & exists	No solution $\exists y$ s.t. $yA \geq 0, yb < 0$
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$\hookrightarrow$ Farkas Alternative

Farkas Alternative: find $y = (y_1, y_2)$ s.t. $y \cdot A \geq 0$, $y \cdot b < 0$

$$y \cdot A = (y_1, y_2) \begin{pmatrix} 2 & 0 & -1 \\ 1 & 1 & 2 \end{pmatrix} = (2y_1 + y_2, y_2, -y_1 + 2y_2) \geq (0, 0, 0)$$

$$y \cdot b = (y_1, y_2) \cdot (1, 0) = y_1 + 0 \cdot y_2 = y_1 < 0$$

$$\begin{cases} 2y_1 + y_2 \geq 0 \\ y_2 \geq 0 \\ -y_1 + 2y_2 \geq 0 \\ y_1 < 0 \end{cases} \Rightarrow (y_1, y_2) \exists ?$$

$$\text{Try } y = (-1, 2) \Rightarrow \begin{cases} 2y_1 + y_2 = -2 + 2 = 0 \quad \checkmark \\ y_2 = 2 > 0 \quad \checkmark \\ -y_1 + 2y_2 = 1 + 2 \cdot 2 = 5 > 0 \quad \checkmark \\ y_1 = -1 < 0 \quad \checkmark \end{cases}$$

$\Rightarrow y = (-1, 2)$ satisfies all four inequalities

$\Rightarrow (-1, 2)$ is a solution of the system

$\Rightarrow (-1, 2)$ is an element of Farkas Alternative

$\Rightarrow$ By Farkas Lemma, the original system $AX = b$ has no non-negative solution.

□

Recap: Span versus cone

Span

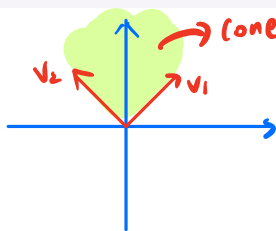
$$\text{span}(\mathbf{A}) = \{\mathbf{y} \in \mathbb{R}^m : \mathbf{y} = \mathbf{A}\mathbf{x} \text{ for some } \mathbf{x} \in \mathbb{R}^n\}.$$

The coefficients in a span may have any sign. *linear combination of columns of A*

Cone

$$\text{cone}(\mathbf{A}) = \{\mathbf{y} \in \mathbb{R}^m : \mathbf{y} = \mathbf{A}\mathbf{x} \text{ for some } \mathbf{x} \in \mathbb{R}_+^n\}.$$

The coefficients in a cone must be non-negative. *non-negative linear combination of columns of A*



$$\mathbf{A}\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0} \iff \mathbf{b} \in \text{cone}(\mathbf{A}).$$

Recap: Comparison of vectors

Componentwise comparisons

For **vectors** $\mathbf{y}, \mathbf{z} \in \mathbb{R}^n$,

$$\mathbf{y} = \mathbf{z} \iff y_i = z_i \text{ for all } i,$$

$$\mathbf{y} \geq \mathbf{z} \iff y_i \geq z_i \text{ for all } i,$$

$$\mathbf{y} > \mathbf{z} \iff y_i \geq z_i \text{ for all } i \text{ and } y_j > z_j \text{ for some } j,$$

$$\mathbf{y} \gg \mathbf{z} \iff y_i > z_i \text{ for all } i.$$

Non-negative orthant

$$\mathbb{R}_+^n \equiv \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} \geq \mathbf{0}\}.$$

Recap: Farkas Lemma

Farkas Lemma

Alternative Theorem

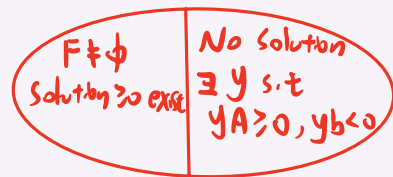
Let $\mathbf{A}$ be an $m \times n$ matrix, $\mathbf{b} \in \mathbb{R}^m$, and

$$F = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{Ax} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}.$$

Exactly one of the following holds:

1. $F \neq \emptyset$;
2. there exists a **vector** $\mathbf{y} \in \mathbb{R}^m$ such that

$$\mathbf{yA} \geq \mathbf{0}, \quad \mathbf{y} \cdot \mathbf{b} < 0.$$



The second system is called the **Farkas alternative**.

Problem 1: Knowledge used

Knowledge used

- Cone generated by the columns of a **matrix**:

$$\text{cone}(\mathbf{A}) = \{\mathbf{Ax} : \mathbf{x} \geq \mathbf{0}\}.$$

- A column that is itself a non-negative linear combination of other columns does not enlarge the cone.
- Farkas Lemma: exactly one of

$$\mathbf{Ax} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}$$

and

$$\mathbf{yA} \geq \mathbf{0}, \mathbf{y} \cdot \mathbf{b} < 0$$

has a solution.

Problem 1(i): Cone generated by the columns

The columns are

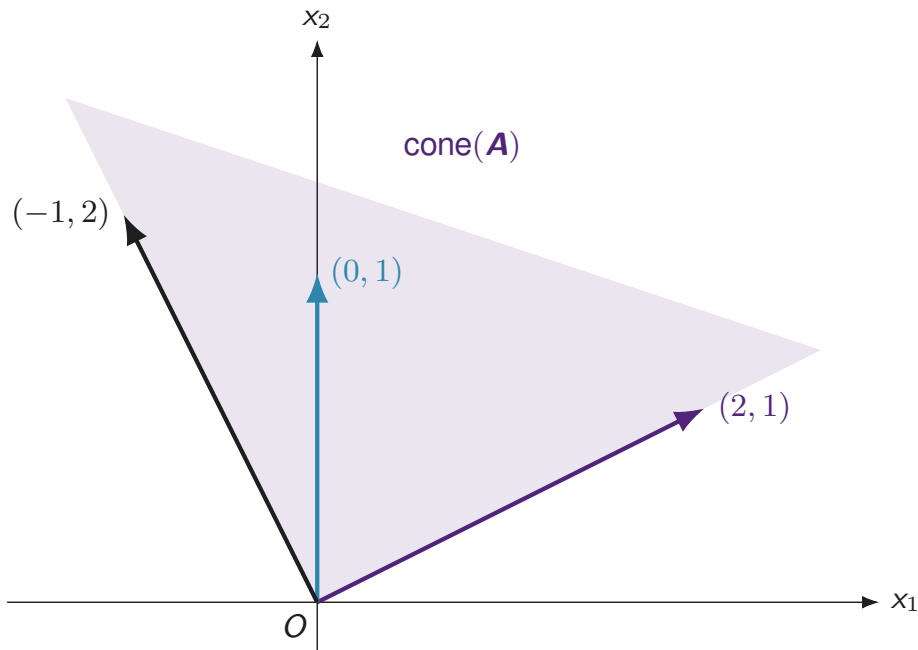
$$\mathbf{a}^1 = (2, 1), \quad \mathbf{a}^2 = (0, 1), \quad \mathbf{a}^3 = (-1, 2).$$

Hence

$$\text{cone}(\mathbf{A}) = \{x_1 \mathbf{a}^1 + x_2 \mathbf{a}^2 + x_3 \mathbf{a}^3 : x_1, x_2, x_3 \geq 0\}.$$

The two extreme rays are generated by $\mathbf{a}^1$ and $\mathbf{a}^3$; the second column lies between them.

Problem 1(i): Sketch



Problem 1(ii): First and third columns

Solution

The cone generated by the first and third columns is the same as the cone generated by all three columns.

Indeed, the second column is a non-negative linear combination of the first and third columns:

$$\frac{1}{5}(2, 1) + \frac{2}{5}(-1, 2) = (0, 1).$$

Therefore,

$$\mathbf{a}^2 \in \text{cone}\{\mathbf{a}^1, \mathbf{a}^3\},$$

and including $\mathbf{a}^2$ does not enlarge the cone.

Problem 1(iii): Farkas alternative

For $\mathbf{b} = (1, 0)$, the Farkas alternative is

$$\mathbf{y}\mathbf{A} \geq \mathbf{0}, \quad \mathbf{y} \cdot \mathbf{b} < 0.$$

Writing $\mathbf{y} = (y_1, y_2)$,

$$(y_1, y_2) \begin{pmatrix} 2 & 0 & -1 \\ 1 & 1 & 2 \end{pmatrix} \geq (0, 0, 0), \quad y_1 < 0.$$

Thus the alternative is

$$\begin{cases} 2y_1 + y_2 \geq 0, \\ y_2 \geq 0, \\ -y_1 + 2y_2 \geq 0, \\ y_1 < 0. \end{cases}$$

Problem 1(iii): Conclusion

Solution

The feasible region of the Farkas alternative is non-empty. For example,

$$\mathbf{y} = (-1, 2)$$

satisfies

$$2(-1) + 2 = 0, \quad 2 \geq 0, \quad -(-1) + 2(2) = 5 \geq 0,$$

and $y_1 = -1 < 0$.

Hence the Farkas alternative has a solution. By Farkas Lemma, the original system

$$\mathbf{Ax} = \mathbf{b}, \quad \mathbf{x} \geq \mathbf{0}$$

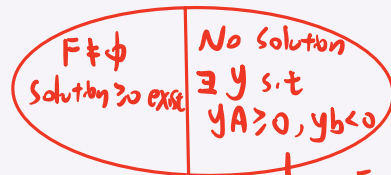
does not have a non-negative solution.

Tutorial Question 2

Problem 2

Use Farkas Lemma to decide whether the system

$$\begin{pmatrix} 4 & 1 & -5 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$



has a non-negative solution.

$$A = \begin{pmatrix} 4 & 1 & -5 \\ 1 & 0 & 2 \end{pmatrix}, b = (1, 1), x = ? \Rightarrow \text{system } Ax = b$$

Farkas Alternative: find $y = (y_1, y_2)$ s.t. $yA \geq 0, yb < 0$

$$yA = (y_1, y_2) \begin{pmatrix} 4 & 1 & -5 \\ 1 & 0 & 2 \end{pmatrix} = (4y_1 + y_2, y_1, -5y_1 + 2y_2) \geq (0, 0, 0)$$

$$yb = (y_1, y_2) \cdot (1, 1) = y_1 \cdot 1 + y_2 \cdot 1 = y_1 + y_2 < 0$$

$$\Rightarrow \text{system } \begin{cases} 4y_1 + y_2 \geq 0 & (1) \\ y_1 \geq 0 & (2) \\ -5y_1 + 2y_2 \geq 0 & (3) \\ y_1 + y_2 < 0 & (4) \end{cases}$$

$\Rightarrow$ determine where the system is empty or not

$$(2) + (4): \quad \begin{array}{cc} y_1 & + & y_2 & < & 0 \\ \underline{\geq 0} & & \underline{< 0} & & \end{array} \Rightarrow y_2 < 0$$

$$(2) + (3): \quad \begin{array}{cc} -5y_1 & + & 2y_2 & \geq & 0 \\ \underline{\leq 0} & & \underline{> 0} & & \end{array} \Rightarrow y_2 > 0$$

$\updownarrow$ contradiction

$\Rightarrow$ the system of these four inequalities has no solution

$\Rightarrow$ Farkas Alternative is empty

$\Rightarrow$ By Farkas lemma, the original system $AX=b$ has a non-negative solution x .

□

Recap: Farkas Lemma

Farkas Lemma

Let $\mathbf{A}$ be an $m \times n$ matrix, $\mathbf{b} \in \mathbb{R}^m$, and

$$F = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{Ax} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}.$$

Exactly one of the following holds:

1. $F \neq \emptyset$;
2. there exists a **vector** $\mathbf{y} \in \mathbb{R}^m$ such that

$$\mathbf{yA} \geq \mathbf{0}, \quad \mathbf{y} \cdot \mathbf{b} < 0.$$

The second system is called the **Farkas alternative**.

Problem 2: Knowledge used

Knowledge used

- Farkas Lemma for

$$\mathbf{Ax} = \mathbf{b}, \quad \mathbf{x} \geq \mathbf{0}.$$

- The Farkas alternative is

$$\mathbf{yA} \geq \mathbf{0}, \quad \mathbf{y} \cdot \mathbf{b} < 0.$$

- To show that the original system has a non-negative solution, show that the Farkas alternative has no solution.

Problem 2: Set up the Farkas alternative

Let

$$\mathbf{A} = \begin{pmatrix} 4 & 1 & -5 \\ 1 & 0 & 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

For $\mathbf{y} = (y_1, y_2)$, the alternative is

$$(y_1, y_2)\mathbf{A} \geq (0, 0, 0), \quad y_1 + y_2 < 0.$$

Equivalently,

$$\begin{cases} 4y_1 + y_2 \geq 0, \\ y_1 \geq 0, \\ -5y_1 + 2y_2 \geq 0, \\ y_1 + y_2 < 0. \end{cases}$$

Problem 2: Rule out the alternative

Solution

From the second inequality,

$$y_1 \geq 0.$$

Together with $y_1 + y_2 < 0$, this implies

$$y_2 < 0.$$

But then

$$-5y_1 + 2y_2 < 0,$$

which contradicts the third inequality

$$-5y_1 + 2y_2 \geq 0.$$

Therefore, the Farkas alternative has no solution.

By Farkas Lemma, the original system **has a non-negative solution**.

Tutorial Question 3

Problem 3

Let $\mathbf{A}$ be an $m \times n$ matrix, and let $\mathbf{b} \in \mathbb{R}^m$. Prove, using Farkas Lemma, that exactly one of the following holds:

1. the system

$$\mathbf{Ax} \geq \mathbf{b}$$

has a non-negative solution $\mathbf{x} \geq \mathbf{0}$;

2. there exists a non-negative vector $\mathbf{y} \in \mathbb{R}^m$ such that

$$\mathbf{yA} \leq \mathbf{0}, \quad \mathbf{y} \cdot \mathbf{b} > 0.$$

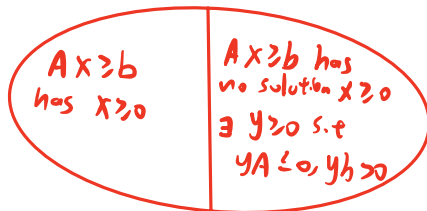
Recap:

$$\mathbf{Ax} \geq \mathbf{b} \xrightarrow{s} \mathbf{Ax} - \mathbf{s} = \mathbf{b}$$

why? $\mathbf{Ax} - \mathbf{b} \geq \mathbf{0}$

Let $\mathbf{s} = \mathbf{Ax} - \mathbf{b} \geq \mathbf{0}$

$$\Rightarrow \mathbf{Ax} - \mathbf{s} = \mathbf{b}$$



WTS

proof: In order to use Farkos Lemma, we have to convert inequality to equality.

Start from $AX \geq b$, and $x \geq 0$

We can always find $s = Ax - b$ such that $AX - s = b$.

$$\Leftrightarrow AX - I \cdot s = b \quad \text{where } I = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix} \text{ is an identity matrix}$$

$$\Leftrightarrow AX + (-I) \cdot s = b \quad \text{inner product: sum} \Leftrightarrow \text{inner product}$$

$$\Leftrightarrow \begin{pmatrix} A & -I \end{pmatrix} \begin{pmatrix} x \\ s \end{pmatrix} = b$$

Augmented matrix
 $\Rightarrow$ Bigger matrix $\mathbb{R}^{m \times (n+m)}$

$$\Leftrightarrow B \cdot z = b, \text{ where } B = (A | -I), z = \begin{pmatrix} x \\ s \end{pmatrix} \geq 0 \quad \left. \begin{matrix} x \geq 0 \\ s \geq 0 \end{matrix} \right\}$$

Farkos Lemma: one of them must be true

- ① $Bz = b$ has a solution $z \geq 0 \Leftrightarrow Ax \geq b$ with $x \geq 0$
- ② $\exists y'$ such that $y' \cdot B \geq 0, y' \cdot b < 0 \Leftrightarrow yA \leq 0, y \geq 0, yb > 0$

$\updownarrow$ mutually exclusive

Look at the Farkos Alternative

$$\begin{matrix} 1 \times m & m \times n & 1 \times m & m \times m \\ y' \cdot B = y' \cdot (A | -I) = (y' \cdot A | -y' \cdot I) \geq 0 \\ \downarrow & & \downarrow & & (a_1, a_2, \dots, a_n | a_{n+1}, \dots, a_{n+m}) \\ 1 \times m & m \times (n+m) & 1 \times n & 1 \times m & \\ \downarrow & & \downarrow & & \\ 1 \times (n+m) & & 1 \times (n+m) & & \end{matrix}$$

$$\Leftrightarrow y' \cdot A \geq 0, -y' \cdot I \geq 0, y' \cdot b < 0$$

Let $y = -y' \geq 0$

$$\Leftrightarrow y \cdot A \leq 0, y \cdot I \geq 0, y \cdot b > 0$$

QED

Recap: Farkas Lemma

Farkas Lemma

Let $\mathbf{A}$ be an $m \times n$ matrix, $\mathbf{b} \in \mathbb{R}^m$, and

$$F = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{Ax} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}.$$

Exactly one of the following holds:

1. $F \neq \emptyset$;
2. there exists a **vector** $\mathbf{y} \in \mathbb{R}^m$ such that

$$\mathbf{yA} \geq \mathbf{0}, \quad \mathbf{y} \cdot \mathbf{b} < 0.$$

The second system is called the **Farkas alternative**.

Recap: Comparison of vectors

Componentwise comparisons

For **vectors** $\mathbf{y}, \mathbf{z} \in \mathbb{R}^n$,

$$\mathbf{y} = \mathbf{z} \iff y_i = z_i \text{ for all } i,$$

$$\mathbf{y} \geq \mathbf{z} \iff y_i \geq z_i \text{ for all } i,$$

$$\mathbf{y} > \mathbf{z} \iff y_i \geq z_i \text{ for all } i \text{ and } y_j > z_j \text{ for some } j,$$

$$\mathbf{y} \gg \mathbf{z} \iff y_i > z_i \text{ for all } i.$$

Non-negative orthant

$$\mathbb{R}_+^n \equiv \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} \geq \mathbf{0}\}.$$

Recap: Converting linear inequalities

Slack and surplus variables

For a $\leq$ inequality, add a non-negative slack variable:

$$\sum_{j=1}^n a_{ij}x_j \leq b_i \iff \sum_{j=1}^n a_{ij}x_j + s_i = b_i, \quad s_i \geq 0.$$

For a $\geq$ inequality, subtract a non-negative surplus variable:

$$\sum_{j=1}^n a_{ij}x_j \geq b_i \iff \sum_{j=1}^n a_{ij}x_j - s_i = b_i, \quad s_i \geq 0.$$

Problem 3: Knowledge used

Knowledge used

- ▶ A system $\mathbf{Ax} \geq \mathbf{b}$ can be converted to equality form by subtracting a non-negative surplus **vector**.
- ▶ An augmented **matrix** can combine the original variables and auxiliary variables into one equality system.
- ▶ Apply Farkas Lemma to the transformed equality system.
- ▶ A sign change in the Farkas **vector** converts $\mathbf{y}' \leq \mathbf{0}$ into a non-negative **vector**.

Problem 3: Convert the inequalities to equalities

Start from

$$\mathbf{Ax} \geq \mathbf{b}, \quad \mathbf{x} \geq \mathbf{0}.$$

Subtract a non-negative surplus **vector** $\mathbf{s} \geq \mathbf{0}$:

$$\mathbf{Ax} - \mathbf{s} = \mathbf{b}.$$

Using the $m \times m$ identity **matrix** $\mathbf{I}$,

$$\mathbf{Ax} - \mathbf{Is} = \mathbf{b}.$$

Define

$$\mathbf{B} \equiv [\mathbf{A} \mid -\mathbf{I}], \quad \mathbf{z} \equiv (\mathbf{x}, \mathbf{s}).$$

Then the system becomes

$$\mathbf{Bz} = \mathbf{b}, \quad \mathbf{z} \geq \mathbf{0}.$$

Problem 3: Apply Farkas Lemma

Farkas Lemma says that exactly one of

$$Bz = b, \quad z \geq 0$$

and its alternative has a solution.

The alternative is: there exists a **vector** y' such that

$$y'B \geq 0, \quad y' \cdot b < 0.$$

Since $B = [A \mid -I]$,

$$y'B = [y'A \mid -y'] \geq 0.$$

Therefore,

$$y'A \geq 0, \quad y' \leq 0, \quad y' \cdot b < 0.$$

Problem 3: Put the alternative in the required form

Define the non-negative **vector**

$$\mathbf{y} \equiv -\mathbf{y}' \geq \mathbf{0}.$$

Multiplying the remaining inequalities by -1 gives

$$\mathbf{y}\mathbf{A} \leq \mathbf{0}$$

and

$$\mathbf{y} \cdot \mathbf{b} > 0.$$

Hence exactly one of the following holds:

$$\boxed{\mathbf{A}\mathbf{x} \geq \mathbf{b}, \mathbf{x} \geq \mathbf{0}}$$

or

$$\boxed{\mathbf{y}\mathbf{A} \leq \mathbf{0}, \mathbf{y} \geq \mathbf{0}, \mathbf{y} \cdot \mathbf{b} > 0}.$$

This is the required statement.

Let $\mathbf{A}$ be an $m \times n$ **matrix**, and let $\mathbf{b} \in \mathbb{R}^m$. Prove, using Farkas Lemma, that exactly one of the following holds:

Tutorial Question 4

1. the system

$$Ax \geq b$$

Farkas Lemma
for inequality system

has a non-negative solution $x \geq 0$;

2. there exists a non-negative **vector** $y \in \mathbb{R}^m$ such that

$$yA \leq 0, \quad y \cdot b > 0.$$

Farkas alternative

Problem 4

Write down the Farkas alternative for

$$\begin{cases} x_1 + 2x_2 + 3x_3 \leq 5, & (-1) \\ x_1 + 3x_2 - 2x_3 \geq 7, \\ x_1 + x_2 + x_3 \leq 2, \\ x_1 - 2x_2 - 3x_3 \geq 3, \end{cases}$$

with

$$x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 \geq 0.$$

$$\begin{aligned} -x_1 - 2x_2 - 3x_3 &\geq -5 \\ x_1 + 3x_2 - 2x_3 &\geq 7 \\ -x_1 - x_2 - x_3 &\geq -2 \\ x_1 - 2x_2 - 3x_3 &\geq 3 \end{aligned}$$

$$\Leftrightarrow \begin{pmatrix} -1 & -2 & -3 \\ 1 & 3 & -2 \\ -1 & -1 & -1 \\ 1 & -2 & -3 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \geq \begin{pmatrix} -5 \\ 7 \\ -2 \\ 3 \end{pmatrix}$$

$$A \cdot x \geq b \quad \text{with } x \geq 0$$

Let A be an $m \times n$ matrix, and let $b \in \mathbb{R}^m$. Prove, using Farkas Lemma, that exactly one of the following holds:

1. the system

$$Ax \geq b$$

has a non-negative solution $x \geq 0$;

2. there exists a non-negative vector $y \in \mathbb{R}^m$ such that

$$yA \leq 0, \quad y \cdot b > 0.$$

Farkas Lemma
for inequality system

Farkas alternative

By Farkas Lemma for inequality system,

the Farkas Alternative is: $\exists y \geq 0$ such that $yA \leq 0, yb > 0$

Let $y = (y_1, y_2, y_3, y_4)$

$$yA = (y_1, y_2, y_3, y_4) \cdot \begin{pmatrix} -1 & -2 & -3 \\ 1 & 3 & -2 \\ -1 & -1 & -1 \\ 1 & -2 & -3 \end{pmatrix} = \begin{pmatrix} -y_1 + y_2 - y_3 + y_4 \\ -2y_1 + 3y_2 - y_3 - 2y_4 \\ -3y_1 - 2y_2 - y_3 - 3y_4 \end{pmatrix}^T \leq 0$$

$$yb = (y_1, y_2, y_3, y_4) \cdot \begin{pmatrix} -5 \\ 7 \\ -2 \\ 3 \end{pmatrix} = -5y_1 + 7y_2 - 2y_3 + 3y_4 > 0$$

$$\Rightarrow \begin{cases} -y_1 + y_2 - y_3 + y_4 \leq 0 \\ -2y_1 + 3y_2 - y_3 - 2y_4 \leq 0 \\ -3y_1 - 2y_2 - y_3 - 3y_4 \leq 0 \\ -5y_1 + 7y_2 - 2y_3 + 3y_4 > 0 \end{cases}$$

Recap: Converting linear inequalities

Slack and surplus variables

For a $\leq$ inequality, add a non-negative slack variable:

$$\sum_{j=1}^n a_{ij}x_j \leq b_i \iff \sum_{j=1}^n a_{ij}x_j + s_i = b_i, \quad s_i \geq 0.$$

For a $\geq$ inequality, subtract a non-negative surplus variable:

$$\sum_{j=1}^n a_{ij}x_j \geq b_i \iff \sum_{j=1}^n a_{ij}x_j - s_i = b_i, \quad s_i \geq 0.$$

Recap: Farkas alternative for $\mathbf{Ax} \leq \mathbf{b}$

Primal system

$$\mathbf{Ax} \leq \mathbf{b}, \quad \mathbf{x} \geq \mathbf{0}.$$

Farkas alternative

$$\mathbf{yA} \geq \mathbf{0}, \quad \mathbf{y} \geq \mathbf{0}, \quad \mathbf{y} \cdot \mathbf{b} < 0$$

Problem 4: Knowledge used

Knowledge used

- ▶ Rewrite all inequalities in a common $\leq$ form.
- ▶ Express the system as

$$\mathbf{Ax} \leq \mathbf{b}, \quad \mathbf{x} \geq \mathbf{0}.$$

- ▶ From Lecture 6, the Farkas alternative for this system is

$$\mathbf{yA} \geq \mathbf{0}, \quad \mathbf{y} \geq \mathbf{0}, \quad \mathbf{y} \cdot \mathbf{b} < 0.$$

Problem 4: Rewrite the $\geq$ inequalities

The second and fourth inequalities are equivalent to

$$-x_1 - 3x_2 + 2x_3 \leq -7,$$

and

$$-x_1 + 2x_2 + 3x_3 \leq -3.$$

Therefore, with

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix},$$

the system can be written as

$$\mathbf{Ax} \leq \mathbf{b}, \quad \mathbf{x} \geq \mathbf{0}.$$

Problem 4: Matrix form

Here

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -3 & 2 \\ 1 & 1 & 1 \\ -1 & 2 & 3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 5 \\ -7 \\ 2 \\ -3 \end{pmatrix}.$$

Thus the primal system is

$$\begin{pmatrix} 1 & 2 & 3 \\ -1 & -3 & 2 \\ 1 & 1 & 1 \\ -1 & 2 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \leq \begin{pmatrix} 5 \\ -7 \\ 2 \\ -3 \end{pmatrix}, \quad \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \geq \mathbf{0}.$$

Problem 4: Farkas alternative

Solution

For a **vector** $\mathbf{y} = (y_1, y_2, y_3, y_4)$, the Farkas alternative is

$$\mathbf{yA} \geq \mathbf{0}, \quad \mathbf{y} \geq \mathbf{0}, \quad \mathbf{y} \cdot \mathbf{b} < 0.$$

Therefore,

$$(y_1, y_2, y_3, y_4) \begin{pmatrix} 1 & 2 & 3 \\ -1 & -3 & 2 \\ 1 & 1 & 1 \\ -1 & 2 & 3 \end{pmatrix} \geq (0, 0, 0),$$

$$(y_1, y_2, y_3, y_4) \geq (0, 0, 0, 0),$$

and

$$5y_1 - 7y_2 + 2y_3 - 3y_4 < 0.$$